

The background of the slide is a dark blue field filled with a complex, glowing network of thin blue lines and small dots, resembling a molecular structure or a data network. Some areas of the network are more densely packed, creating a sense of depth and connectivity.

Lecture slides of the course
Information hiding & secret sharing

COURSE INTRODUCTION

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Main topics in this course

- Steganography
- Digital watermarking
- Secret sharing

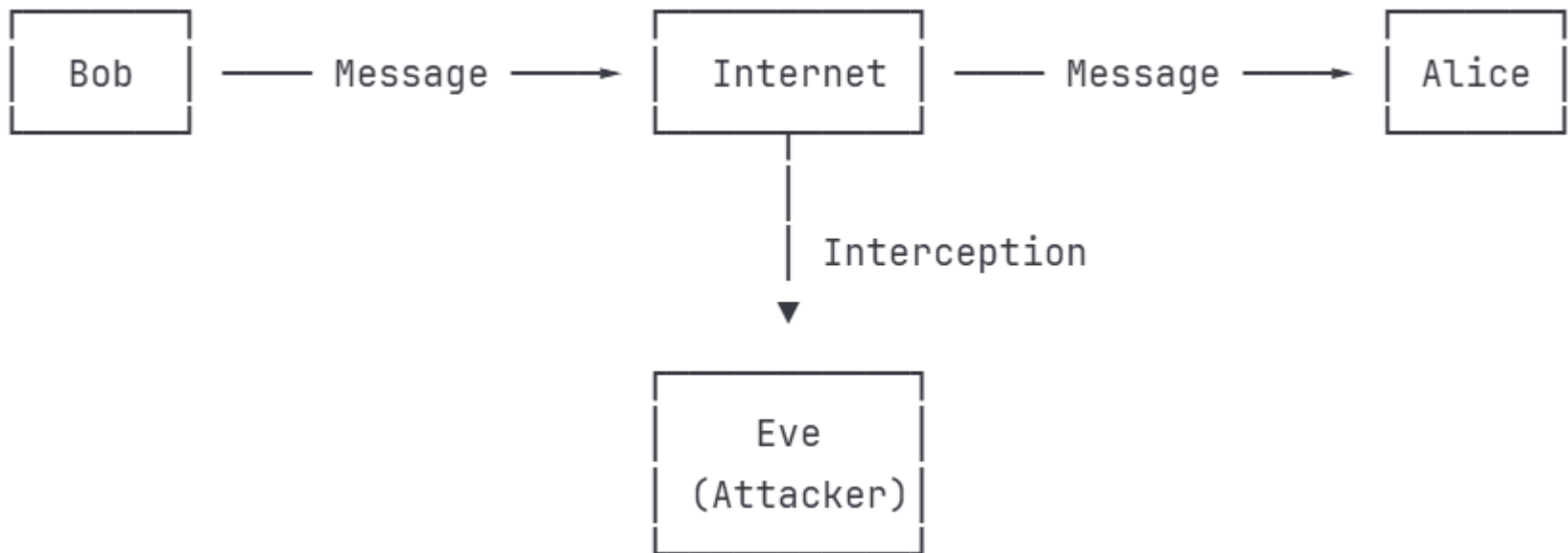
Main topics in this course

- **Steganography**
- Digital watermarking
- Secret sharing

The Need for Secure Communication

The Communication Security Problem

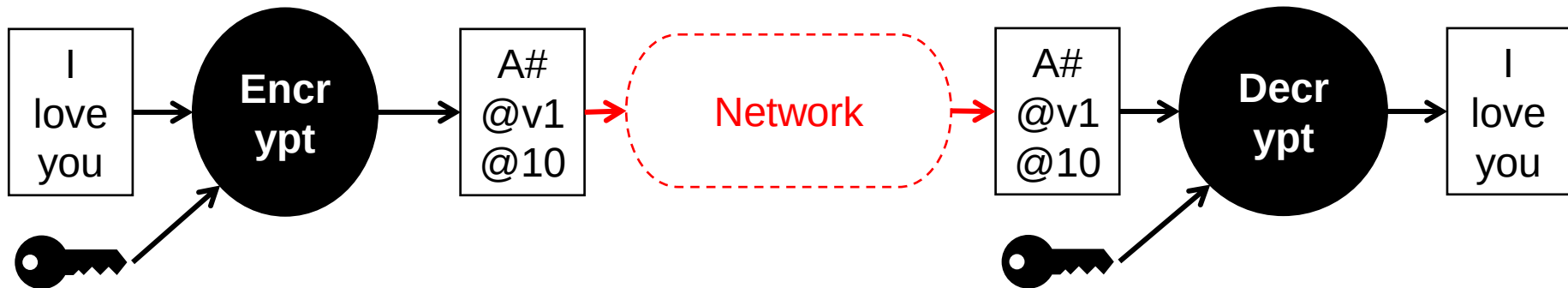
- **Scenario:** Bob wants to send a private message - "I love you." - to Alice over the Internet.
- How can Bob ensure that this information is transmitted securely?



Solution 1: Cryptography (Encryption)

Encrypt the information to be transmitted

The attacker knows the **existence of the info**,
but it is difficult to know the **content of the info**



Aspect	Description
What it hides	The content of the message
What it reveals	The existence of communication
Attacker's knowledge	Knows communication exists, but cannot read content
Security basis	Mathematical complexity of decryption without key

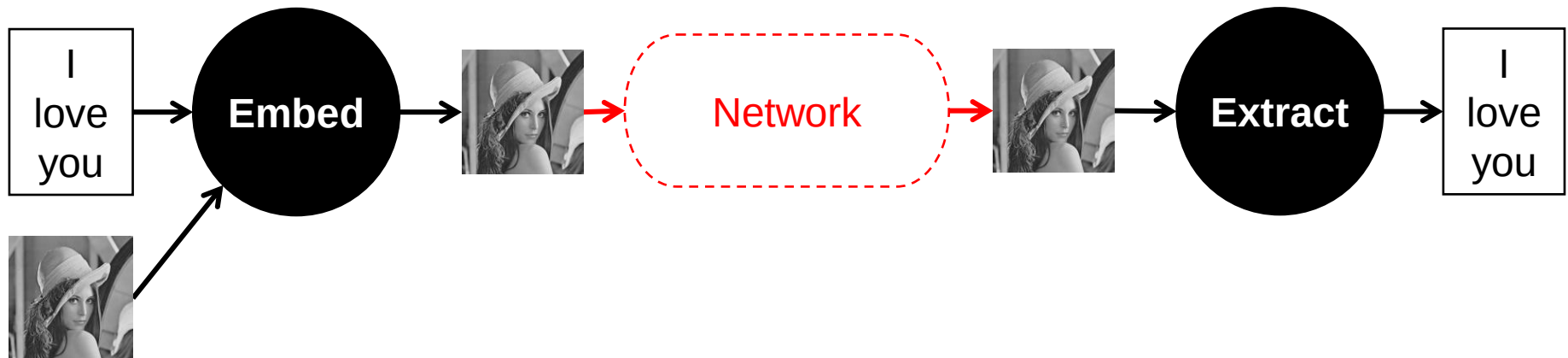
Limitation

- **The attacker knows communication is happening.**
- In some scenarios, merely knowing that two parties are communicating can be valuable intelligence. For example:
 - In authoritarian regimes, encrypted communication may draw suspicion
 - In corporate espionage, knowing competitors are communicating may be significant
 - In military contexts, communication patterns can reveal operational plans

Solution 2: Steganography

Hide the information to be transmitted in another message

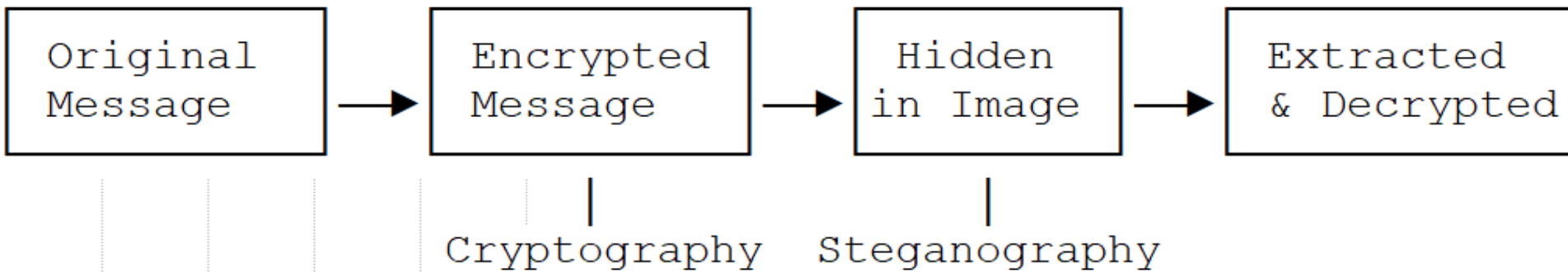
The attacker is **not even aware of the existence of the information**



Aspect	Description
What it hides	The existence of the message itself
What it reveals	Nothing - appears as normal communication
Attacker's knowledge	Unaware that secret communication exists
Security basis	Imperceptibility of hidden data

Solution 3: Combined Approach

- The most secure approach combines both techniques:
 - **First:** Encrypt the secret message (cryptography)
 - **Then:** Hide the encrypted message in a cover medium (steganography)

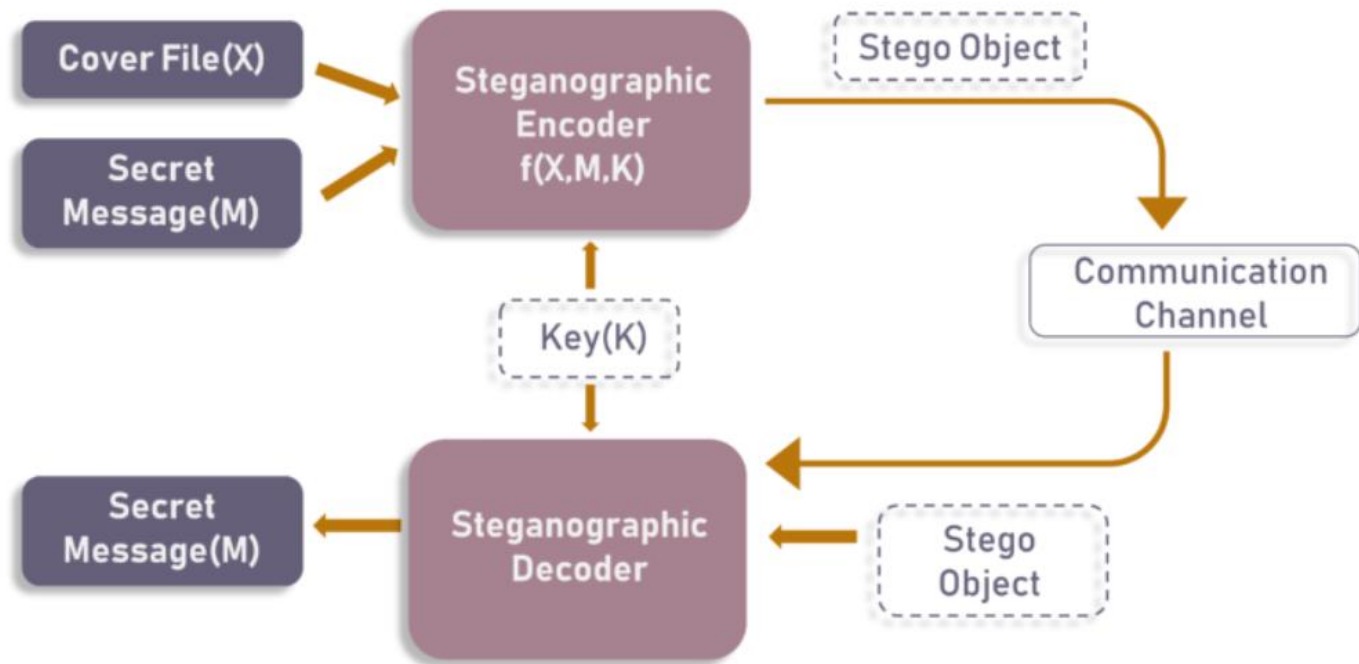


- **Benefits:**
 - **Layer 1 (Steganography):** Attacker doesn't know communication exists
 - **Layer 2 (Cryptography):** Even if hidden data is discovered, it remains unreadable

Steganography - Concepts and Fundamentals

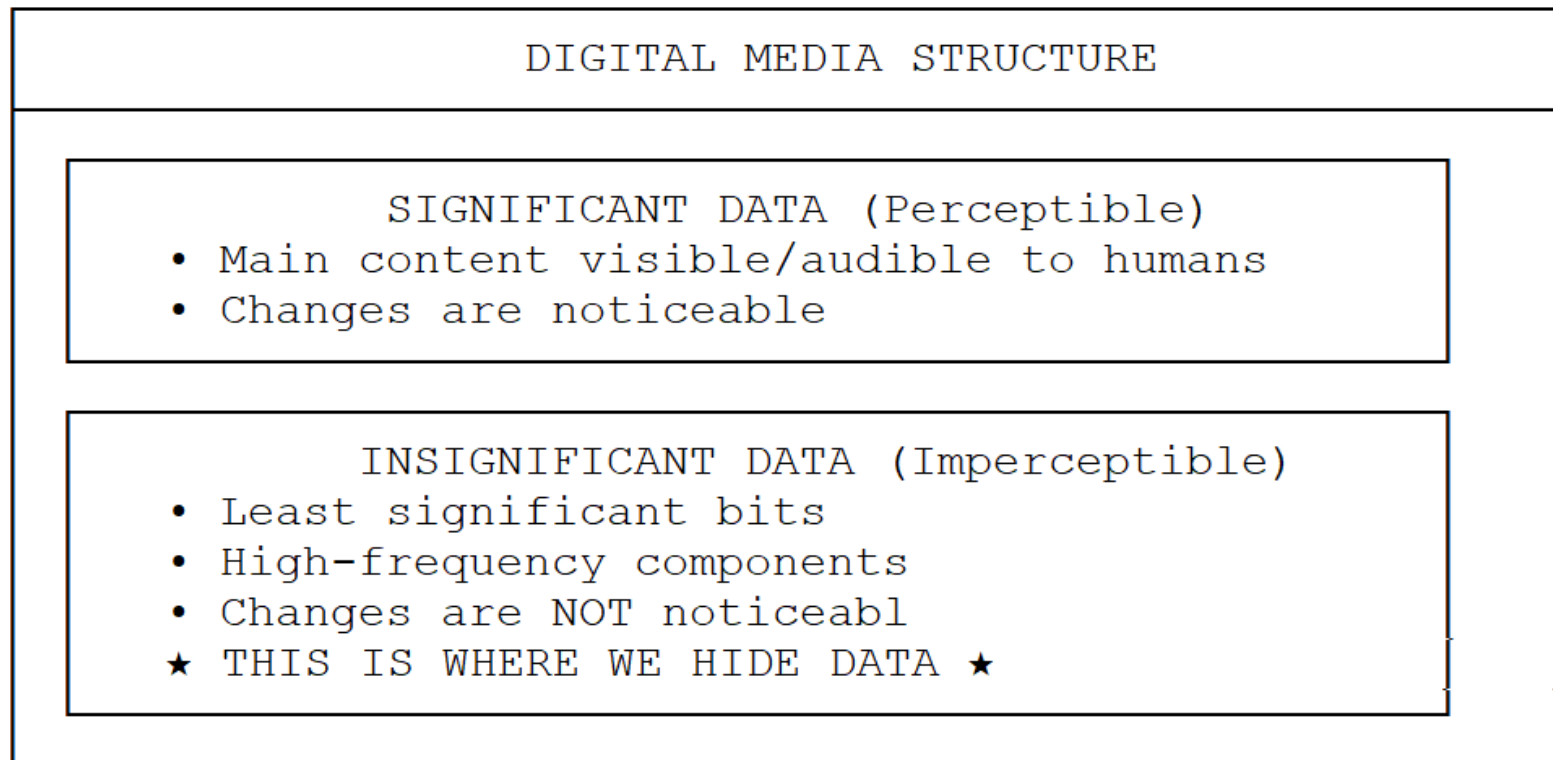
Definition

- **Steganography** is the art and science of **embedding secret messages** in a **cover message** in such a way that no one, apart from the sender and intended recipient, suspects the existence of the message



Core Principle

- Exploits the fact that digital media contains **redundant** or **insignificant** data that can be modified without noticeably affecting the media's appearance or function.



Steganography: Historical Background

- **Steganography** is the practice of concealing a secret message behind a normal message.
- **Etymology:** "steganography" comes from Greek:
 - **Steganos (στεγανός):** means "covered" or "concealed"
 - **Graphia (γραφή):** means "writing"
 - **Steganography:** literally "covered writing"
- Steganography is an ancient practice, being practiced in various forms for thousands of years to keep communications private

Steganography: Historical Background

- The first documented use of steganography traces back to ancient Greece (440 BC):
- **Technique 1: Wax Tablets**
 - Messages were written on wooden tablets
 - The tablets were then covered with wax
 - The wax surface appeared blank or innocent text
 - The recipient would melt the wax to reveal the hidden message

Steganography: Historical Background

- The first documented use of steganography traces back to ancient Greece (440 BC):
- **Technique 2: The Shaved Head Method**
 - Documented by Herodotus in the History of the Persian Wars
 - A lord shaved his messenger's head and tattooed a message on the scalp
 - The messenger waited until hair grew back, concealing the message
 - At the destination, the head was shaved again to reveal the message
 - *Limitation:* Very slow - required weeks for hair to regrow

Steganography: Historical Background

Invisible Inks

- Romans developed various forms of invisible inks
- Messages written with these inks were invisible under normal conditions
- Light or heat was applied to reveal the hidden messages
- Common substances used:
 - Milk
 - Fruit juices (especially lemon)
 - Urine
 - Certain plant extracts

Steganography: Historical Background

(World War II Era)

- **Microdots**

- Developed by German intelligence
- Complete documents, photographs, and plans were photographically reduced to the size of a period (.)
- These microdots were attached to normal paperwork
- Required magnification to read
- Extremely difficult to detect without knowing what to look for

- **Null Ciphers**

- Used to hide unencrypted messages within innocent-looking text
- Secret message extracted using a predetermined rule
- Example: Take the first letter of each word, or every nth letter

Modern Digital Era

- With the advent of computers and digital media, steganography has evolved to exploit:
 - Digital image files (JPEG, PNG, BMP, GIF)
 - Audio files (MP3, WAV, FLAC)
 - Video files (MP4, AVI, MOV)
 - Text documents (PDF, DOC)
 - Network protocols (TCP/IP headers)
 - File systems (slack space, alternate data streams)

Computer steganography

- **Basic Principle**
- Computer steganography takes advantage of the fact that digital files contain unused or insignificant areas of data. These areas can be replaced with secret information without visibly affecting the file.

Steganography vs Cryptography

- Understanding the differences between steganography and cryptography is crucial for knowing when to use each

Aspect	Steganography	Cryptography
Definition	Technique to hide the existence of communication	Technique to convert data into incomprehensible form
Primary Goal	Conceal the fact that communication is occurring	Protect the content of the message
Security Focus	Keep communication secret	Provide data protection
Data Visibility	Hidden data is never visible	Encrypted data is always visible (but unreadable)
Data Structure	Does not alter the overall structure of cover media	Completely alters the structure of data

Steganography vs Cryptography

Aspect	Steganography	Cryptography
Key Requirement	Optional (but recommended for added security)	Absolutely necessary
Failure Mode	If presence is detected, message is easily recovered	Even if intercepted, message remains protected if key is secure
Suspicion Level	Does not raise suspicion	Raises suspicion (obviously encrypted)
Carrier Required	Yes (cover medium needed)	No (standalone encrypted message)

Steganography Techniques by Media Type

Classification Overview

Classification Overview

- Depending on the nature of the cover object, steganography can be divided into five main types:
 - **Text Steganography**
 - **Image Steganography**
 - Video Steganography
 - **Audio Steganography**
 - Network Steganography

Text Steganography

- Text steganography involves hiding information inside text files. This is challenging because text files have very little redundant data compared to multimedia files.
- **Techniques for Text Steganography**
 - Format-Based Methods
 - Random and Statistical Generation
 - Linguistic Methods

Text Steganography

- **Format-Based Methods:** Manipulate text formatting:
 - **Line Shifting:** Slightly shift lines up or down
 - **Word Shifting:** Adjust horizontal spacing between words
 - **Feature Coding:** Modify character features (extending letter strokes)
 - **Whitespace Manipulation:** Use spaces and tabs to encode data
- **Random and Statistical Generation**
 - Generate random-looking character sequences that encode data
 - Create text that appears random but contains hidden structure
- **Linguistic Methods**
 - **Synonym Substitution:** Replace words with synonyms to encode bits
 - **Syntactic Methods:** Modify sentence structure
 - **Semantic Methods:** Change meaning in subtle ways

Text Steganography

- Normal text: "Hello world"
- With hidden data:
 - "Hello world" (extra space encodes '1')
 - "Hello world" (single space encodes '0')

Image Steganography

- Image steganography is the most popular form due to the high redundancy in digital images.
- There are numerous bits in a digital representation of an image that can be modified without visible effect.
- **Techniques:**
 - Least Significant Bit (LSB) Insertion
 - Masking and Filtering
 - Redundant Pattern Encoding
 - Encrypt and Scatter
 - Transform Domain Methods

Image Steganography

- **Least Significant Bit (LSB) Insertion**

- The most common technique. Replaces the least significant bits of pixel values with secret data.

- **Masking and Filtering**

- Hides information in more significant areas
- Like watermarking
- More robust but lower capacity

- **Redundant Pattern Encoding**

- Scatter secret message throughout the image
- Uses patterns to increase robustness

Image Steganography

- **Encrypt and Scatter**

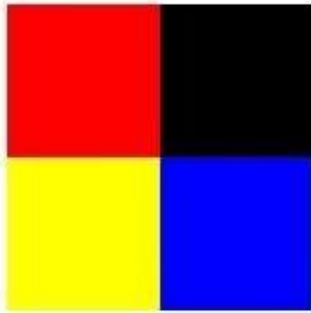
- First encrypt the message
- Then scatter encrypted bits throughout the image

- **Transform Domain Methods**

- **DCT (Discrete Cosine Transform):** Hide in frequency coefficients
- **DWT (Discrete Wavelet Transform):** Hide in wavelet coefficients
- More robust against compression

Image Steganography - Example

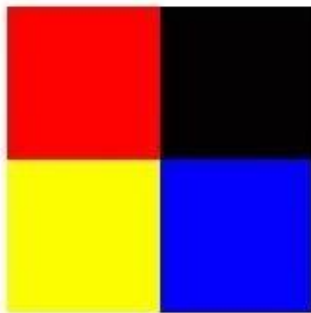
Original Image



11111111	00000000
00000000	00000000
00000000	00000000
11111111	00000000
11111111	00000000
00000000	11111111

Least Significant Bit Steganography

Stego Image



111111 01	000000 11
000000 10	000000 01
000000 00	000000 10
111111 00	000000 11
111111 01	000000 01
000000 01	111111 00



c	a	t
01 10 00 11	01 10 00 01	01 11 01 00

Audio Steganography

- Audio steganography hides data in digital audio files by exploiting the limitations of human hearing (Human Auditory System - HAS).
- **Techniques:**
 - LSB Coding
 - Phase Coding
 - Spread Spectrum
 - Echo Hiding
 - Parity Coding

Audio Steganography

- **LSB Coding**

- Similar to image LSB, modifies least significant bits of audio samples.

- **Phase Coding**

- Modifies the phase of audio segments
- Human ear is less sensitive to phase changes
- More robust than LSB

- **Spread Spectrum**

- Spreads secret data across frequency spectrum
- Uses noise-like signals
- Difficult to detect and remove

- **Echo Hiding**

- Embeds data by introducing echoes
- Controls echo parameters (delay, amplitude, decay)
- Natural-sounding result

Video Steganography

- Video steganography combines techniques from image and audio steganography, with additional temporal dimension.
- **Approaches**
 - Frame-Based
 - Motion Vector
 - DCT Coefficient
- **Challenges**
 - Large file sizes
 - Compression artifacts
 - Temporal consistency
 - Real-time requirements

Network Steganography

- Network steganography hides information in network protocols and traffic patterns.

Techniques

- **Protocol-Based**

- Modify unused or optional header fields
- TCP/IP headers have many opportunities
- ICMP, DNS, HTTP headers

- **Timing-Based**

- Encode data in packet timing
- Inter-packet delays carry information
- Difficult to detect

- **Storage-Based**

- Use covert channels in network protocols
- Exploit protocol ambiguities

Steganography Examples

Toy Example 1: Capital Letter Cipher

- This example demonstrates a simple text steganography technique using capital letters.
- The Cover Message

randoM capitalosis is a rarE disEase ofTen contrAcTed by careless inTernet users. tHis sad illnEss causes the aFfected peRsON To randomly capitalize letters in a bOdy oF texT. please do not confuse this disease with a blatant attEmpt aT steganogRAPhy.

Toy Example 1: Capital Letter Cipher

- The Cover Message

randoM capitalosis is a rarE disEase ofTen contrAcTed by careless inTernet users. tHis sad illnEss causes the aFfected peRsON To randomly capitalize letters in a bOdy oF texT. please do not confuse this disease with a blatant attEmpt aT steganogRAPhy.

- **Extraction Process**

- **Step 1:** Extract all capital letters that appear unusually in the middle of words:

- M E E T A T T H E F R O N T O F T H E T R A P

- **Step 2:** Read the message:

- MEET AT THE FRONT OF THE TRAP

Toy Example 2: First Letter + Mathematical Cipher

- This more sophisticated example combines multiple techniques.
- Alice sends this email to Bob:

“My friend Bob,
Until yesterday I was using binoculars for stargazing. Today, I decided to try my new telescope. The galaxies in Leo and Ursa Major were unbelievable! Next, I plan to check out some nebulas and then prepare to take a few snapshots of the new comet. Although I am satisfied with the telescope, I think I need to purchase light pollution filters to block the xenon lights from a nearby highway to improve the quality of my pictures.
Cheers,
Alice”

Toy Example 2: First Letter + Mathematical Cipher

“My friend Bob,

Until yesterday I was using binoculars for stargazing. Today, I decided to try my new telescope. The galaxies in Leo and Ursa Major were unbelievable! Next, I plan to check out some nebulas and then prepare to take a few snapshots of the new comet. Although I am satisfied with the telescope, I think I need to purchase light pollution filters to block the xenon lights from a nearby highway to improve the quality of my pictures.

Cheers,

Alice”

Step 1: Listing the first letters of all words

**mfbuyiwubfstidttmnttgilaumwuniptcosnatpttafsotncaiaswttitintplpftbt
xlfanhtitqompca.**

Toy Example 2: First Letter + Mathematical Cipher

Step 1: Listing the first letters of all words

**mfbuyiwubfstidttmnttgilaumwuniptcosnatpttafsotncaiaswttitintplpftbt
xlfanhtitqompca.**

Step 2: $\pi = 3.141592653589793$

Reads the message from the extracted sequence of letters by putting down the third letter in the sequence, then the next first letter, the next fourth letter, etc. The resulting message is:

buubdlupnpsspx

Toy Example 2: First Letter + Mathematical Cipher

Step 1:

mfbuyiwubfstidttmnttgilaumwuniptcosnatpttafsotncaiaswttitintplpftbt
xlfanhtitqompca.

Step 2: buubdlupnpssp

Step 3: replaces each letter with the letter that precedes it in the alphabet
and deciphers the secret message

attack tomorrow.

Why Steganalysis Matters

- While steganography has many legitimate uses, it can also be misused for:
- **Terrorism:** Hidden messages in innocent-looking media can coordinate attacks without detection. This became a major concern after September 11th, 2001.
- **Espionage:** Spies use steganography to pass messages through public channels. The Anna Chapman Russian spy ring, for example, hid messages in images on public websites.
- **Cybercrime:** Attackers use steganography to hide malware in images or audio files, exfiltrate stolen data in outgoing files, and embed botnet commands in web content. Researchers have demonstrated that malicious code hidden in images can evade sophisticated malware scanners.

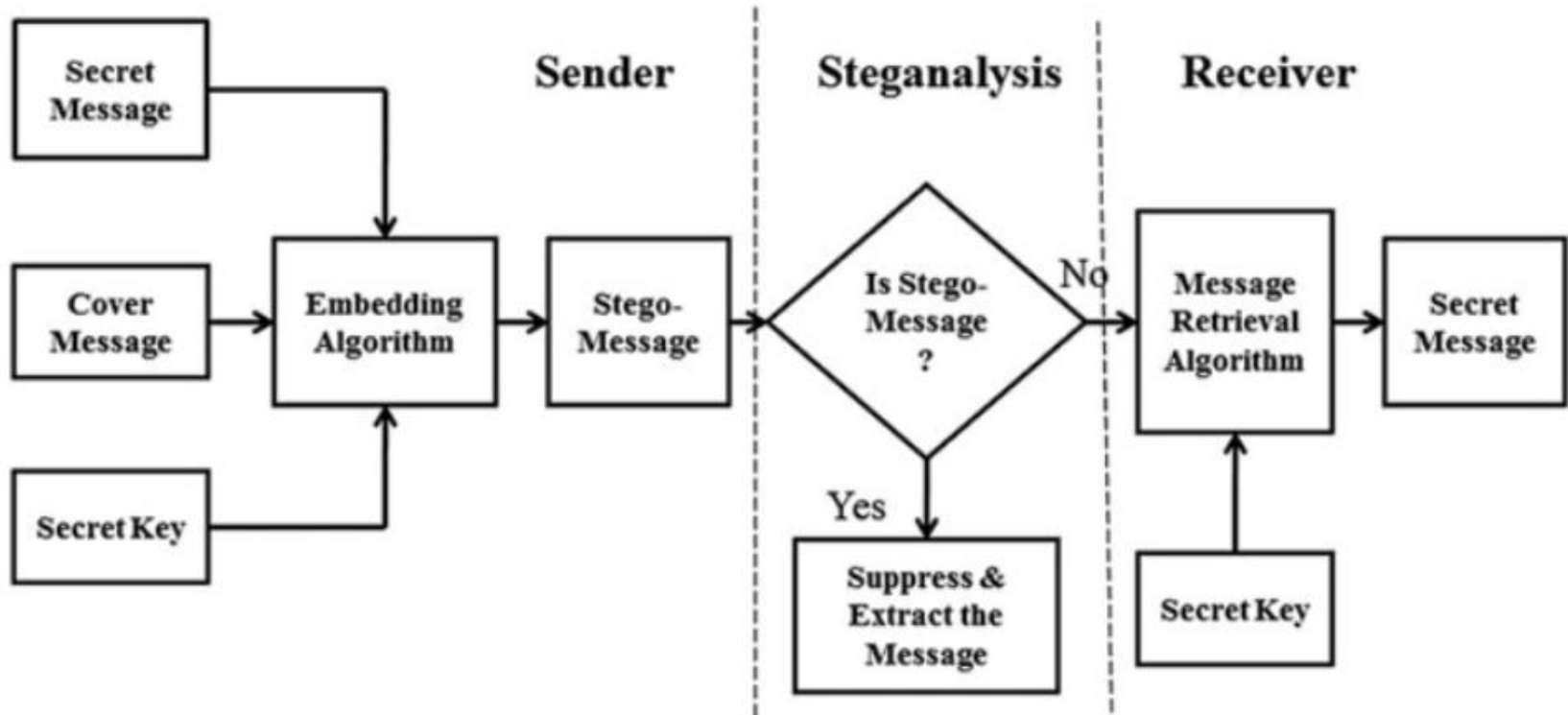
Steganalysis

- **Steganalysis:** the set of techniques and methods used to:
 - **Detect** the presence of information embedded in digital objects
 - **Extract** this information if it is possible
- Steganalysis involves **analysis** of the carrier file

Steganalysis Techniques

- **Visual Analysis**
 - Examine images for visual anomalies
 - Look for unusual patterns in bit planes
 - Check for distortions visible to trained eye
- **Statistical Analysis:** Steganography often leaves statistical fingerprints:
 - **Histogram Analysis:** Check for unusual patterns in value distributions
 - **Chi-Square Attack:** Detect pairs of values created by LSB replacement
 - **RS Analysis:** Identify regular and singular pixel groups
 - **Sample Pair Analysis:** Advanced statistical detection

Steganalysis



Steganalysis - Classification

Specific (Targeted) Steganalysis

When the steganographic algorithm is **known**:

- Detector is designed specifically for that algorithm
- Analyzes statistical properties that change after embedding
- **Advantage:** Very accurate results
- **Disadvantage:** Limited to specific algorithms and formats

Steganalysis - Classification

Blind (Universal) Steganalysis

When the steganographic algorithm is **unknown**:

- Detector works regardless of embedding method
- Uses machine learning and general statistical features
- **Advantage:** Works against unknown methods
- **Disadvantage:** Less accurate than targeted approaches

Comparison

Aspect	Specific Steganalysis	Blind Steganalysis
Algorithm knowledge	Required	Not required
Accuracy	High	Moderate
Generalization	Low	High
Current research focus	Declining	Increasing
Practical use	Limited	Widespread

Steganalysis tools

Tool	Purpose	Platform
StegDetect	Detect steganography in JPEG images	Cross-platform
StirMark	Benchmark watermarking robustness	Windows
StegBreak	Dictionary attack on steganography	Linux
StegSecret	Detect and extract hidden data	Cross-platform
JPSeek	Analyze JPEG images	Cross-platform
2Mosaic	Visual steganalysis	Windows

Main topics in this course

- Steganography
- **Digital watermarking**
- Secret sharing

The need for copyright protection

- **Problem Scenario 1: Photo Theft**
- You take a beautiful photograph and share it on the internet. Others can easily:
 - Copy your photo
 - Claim it as their own
 - Use it for commercial purposes without permission
 - Remove any visible attribution
- **How do you prove the photo is yours?**

The need for copyright protection

- **Problem Scenario 2: Content Leakage**
- You are an e-book publisher selling digital copies to customers. A customer can:
 - Create unlimited copies
 - Share with others
 - Upload to piracy websites
 - Result: Your content is freely available, destroying your business
- **How do you identify the source of the leak?**

Solution: Digital Watermarking

- A **digital watermark** is a pattern of bits inserted into a digital media file that can identify the creator or authorized users.



The Three Criteria Triangle

- Every data hiding system must balance three competing criteria:

Criterion	Definition	Measurement
Robustness	Ability of hidden message to remain undamaged after transformations (scaling, cropping, compression, etc.)	Bit Error Rate (BER) after attacks
Invisibility	Similarity between stego/watermarked media and original	PSNR, SSIM
Capacity	Amount of data that can be hidden	Bits per pixel (bpp)

Steganography vs watermarking

Among the 3 criteria: invisibility - robustness – capacity

- Steganography often places heavy emphasis on **invisibility** and **capacity**
- Watermarking often places heavy emphasis on **robust** and **invisibility**

Criterion	Steganography Priority	Watermarking Priority
Invisibility	★★★★★ (Critical)	★★★★☆ (Very Important)
Capacity	★★★★☆ (Important)	★★☆☆☆ (Less Important)
Robustness	★★☆☆☆ (Secondary)	★★★★★ (Critical)

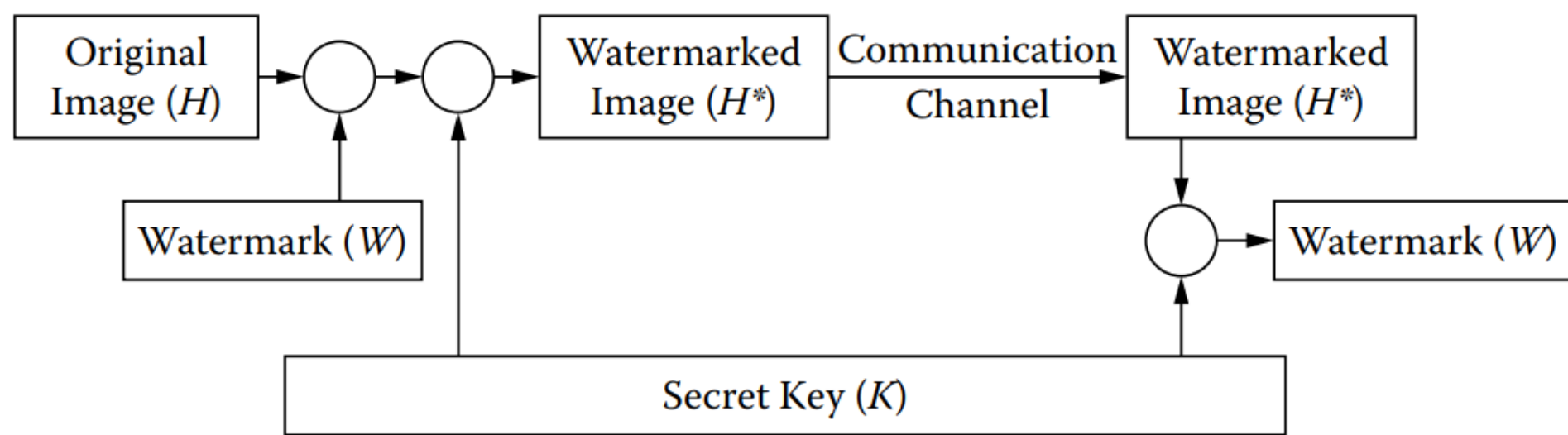
Steganography vs watermarking

Aspect	Steganography	Watermarking
Primary Purpose	Covert communication	Copyright protection
Message Relationship	Secret message unrelated to cover	Watermark relates to cover content
Detection Goal	Avoid all detection	Survive attacks, remain detectable
Typical Payload	Large (documents, files)	Small (IDs, signatures)
Robustness Needs	Low (just deliver once)	High (must survive processing)
Key Concern	Existence discovery	Watermark removal

Requirements for Ideal Watermarking

- **Imperceptibility:** Must be indistinguishable from original
- **Large Capacity:** Embed sufficient identification data
- **Perfect Security:** Unauthorized parties cannot detect/remove
- **No Degradation:** Host media quality maintained
- **Robustness:** Survive intentional attacks (noise, cropping) and unintentional processing (compression, enhancement)

General digital watermarking system



Main topics in this course

- Steganography
- Digital watermarking
- **Secret sharing**

The Secret Sharing Problem

- You want to store secret information (password, encryption key, rocket launch code, etc.). You have two requirements:
 - **Confidentiality**: Information must not be known by unauthorized parties
 - **Availability**: Information must not be lost

The Secret Sharing Problem

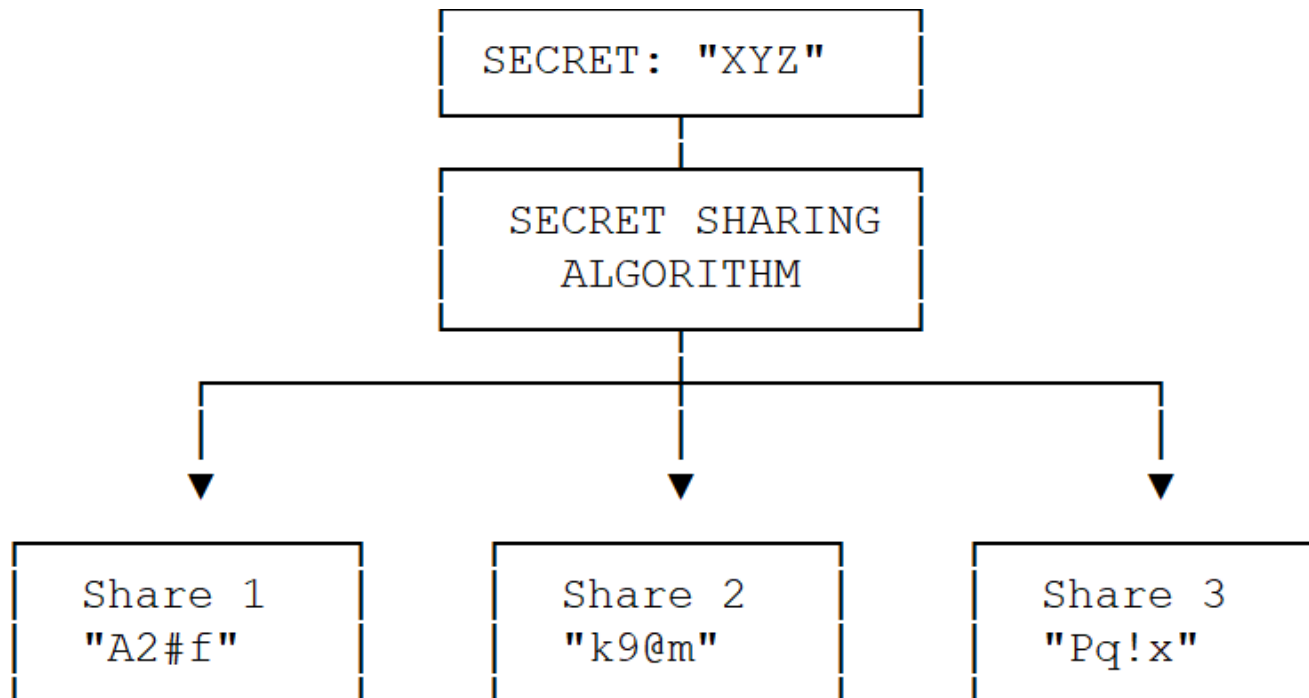
- **Solution 1: Single Copy Storage**
- Store one copy in one secure location (e.g., a computer).
- **Advantages:**
 -  Difficult for others to find (only one location to protect)
- **Disadvantages:**
 -  Single point of failure (if computer fails, secret is lost forever)
 -  No backup or recovery option

The Secret Sharing Problem

- **Solution 2: Multiple Copy Storage**
- Store identical copies in multiple locations.
- **Advantages:**
 -  Redundancy (multiple copies = lower loss probability)
- **Disadvantages:**
 -  Increased attack surface (attacker only needs to compromise ONE location)
 -  More copies = more risk of exposure

The Secret Sharing Problem

- **Solution 3: Secret Sharing**
- Divide the secret into **n parts** (called **shares**), stored in n different locations. To recover the original secret, at least **k parts** are required (where $k \leq n$).



The Secret Storage Problem

Threshold Scheme (k, n)

- A **(k, n) threshold scheme** divides secret S into n shares such that:
 - Any **k or more** shares can reconstruct S
 - Any **fewer than k** shares reveal **nothing** about S

Property	Description
Availability	Can lose up to $(n-k)$ shares and still recover
Confidentiality	Attacker needs at least k shares
Flexibility	Can choose k and n based on needs

The need for secret sharing

Sharing secret information can also be used when there is a need to share a secret information with a group of people

- We wish that only a certain number of people would be able to recover secret information
- We also expect that if a certain number of people have a problem, it is still possible to recover secret information from the rest.

Scientific and Commercial Applications

The Growing Importance of Data Security

- **Digital data explosion**
- Every day, the world generates:
 - 500 million tweets
 - 294 billion emails
 - 4 petabytes of Facebook data
 - 720,000 hours of YouTube videos
- Data **breaches** exposed:
 - 22 billion records in 2021
 - Average cost: \$4.45 million per breach (2023)
- Cybercrime **costs**:
 - \$8 trillion globally in 2023
 - ~\$10.5 trillion by 2025

Cryptography Alone Is Not Enough

- You might think: *"I already know about encryption. Isn't that enough?"*. The answer is **NO**. Here's why:

Scenario	Cryptography	Information Hiding	Best Solution
Sending secrets in monitored network	Encrypted traffic is flagged	Hidden in normal traffic	Steganography
Proving you own a digital photo	Cannot help	Embedded watermark	Watermarking
Storing master password securely	Single point of failure	Distributed shares	Secret Sharing
Passing censorship firewalls	Blocked or detected	Looks like normal data	Steganography
Tracking leaked documents	Cannot help	Hidden fingerprint	Watermarking

Cryptography Alone Is Not Enough

- You might think: *"I already know about encryption. Isn't that enough?"*. The answer is **NO**.
- **Information hiding complements cryptography** - they work together to provide complete security solutions.

Real-World Applications You'll Encounter

- Streaming Services (Netflix, Spotify, YouTube)
- When you stream a movie:



- If you record and leak it → Watermark identifies YOU
- Netflix, Disney+, and others use forensic watermarking to track the source of pirated content

Real-World Applications You'll Encounter

- **Social Media Platforms**

- **Facebook/Instagram:** Use perceptual hashing to detect copyrighted content
- **TikTok:** Embeds invisible watermarks in downloaded videos
- **Twitter/X:** Strips metadata but platforms use content fingerprinting

- **Currency and Documents**

- **Banknotes:** Contain multiple hidden security features
- **Passports:** Use digital watermarks and hidden data
- **ID Cards:** Embed verification data invisibly

- **Music Industry**

- **Audio watermarks** identify songs on radio/TV for royalty tracking
- **Shazam/SoundHound** use audio fingerprinting (related technology)

Real-World Applications You'll Encounter

- **Medical Imagery**
- Hospitals and clinical doctors use steganography to embed patient information directly in medical images.

Benefit	Description
Data Integrity	Patient info travels with image
Error Reduction	Reduces mismatch between image and patient record
Fraud Prevention	Difficult to swap or modify records
Privacy	Information hidden from casual viewing

Medical-image steganography requires extreme care. The additional information must NOT affect diagnostic image quality.

Real-World Applications You'll Encounter

- **Banking & Finance**

- **Transaction watermarking** for audit trails
- **Document authentication** for contracts
- **Secret sharing** for cryptographic key management

- **Military & Intelligence**

- **Covert communication** channels
- **Authentication** of orders and documents
- **Satellite imagery** protection

- **Legal & Forensics**

- **Evidence authentication** (proving photos/videos weren't tampered)
- **Chain of custody** verification
- **Digital forensics** (detecting hidden data in seized devices)

Career Opportunities

- Studying this course prepares you for various career paths
- **Cybersecurity Specialist**
 - Implement security measures including steganography detection
- **Digital Forensics Analyst**
 - Detect hidden data in criminal investigations
- **Media Security Engineer**
 - Develop watermarking systems for content protection
- **Research Scientist**
 - Develop new steganography/steganalysis algorithms
- **Blockchain/Cryptocurrency Developer**
 - Implement secret sharing for key management

Industries Hiring These Skills

Industry	How They Use Information Hiding
Entertainment	Content protection, piracy tracking
Healthcare	Medical image security, patient data
Finance	Document authentication, secure key storage
Government	National security, document verification
Technology	Platform security, content moderation
Legal	Digital forensics, evidence verification
Defense	Covert communication, intelligence
Journalism	Source protection, secure communication

Real-World Impact

- **Detect deepfakes** and AI-generated misinformation
- **Protect creators'** intellectual property
- **Enable secure communication** for journalists and activists
- **Support law enforcement** in digital investigations

Skills You Will Gain

- Technical Skills

Skill Area	What You'll Learn
Programming	Implement steganography & watermarking algorithms
Signal Processing	DCT, DWT, frequency domain analysis
Machine Learning	Steganalysis using classifiers & deep learning
Statistical Analysis	Hypothesis testing, anomaly detection

Skills You Will Gain

- Transferable Skills

Skill	Broader Application
Algorithm Design	Any software development
Security Mindset	All cybersecurity roles
Signal Processing	Image/video/audio processing, computer vision
Statistical Analysis	Data science, machine learning
Research Methods	Graduate studies, R&D positions

Course assessment

- **Labs: (30%)**
 - Programing language: Python
 - Working on IPython Notebook
- **Seminar (30%)**
 - Group work, 2 student each group
 - Research on data hiding topics and give oral presentation
- **Final exam (40%)**
 - Written and MCQ

Course regulations

- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.
- Students absent for final exam are considered unqualified for course completion.
- Students must accumulate at least 10% of course credits for lab work.

References

- <https://medium.com/@rabi3elbeji/friendly-introduction-to-steganography-4cf032240904>